

Capstone Project (WK7)

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1.0 Abstract

This study investigates the impact of demographic variables such as age, education, and marital status on the success of Certificate Deposit (CD) marketing campaigns in the banking industry. Using the Kaggle dataset of 45,211 records from a Portuguese banks telemarketing data, the analysis of descriptive statistics, correlation analysis, and predictive modeling was used to test the null hypothesis that these demographic factors do not significantly influence campaign outcomes.

The main findings were that Duration had the strongest correlation of successful outcomes. This indicated a strong relationship between longer call durations and positive outcomes. Secondly, Education seemed to play a significant role as a predictor as well. With customers who hold a college level education, they had the highest success rates. Analysis of marital status also suggested that divorced individuals had a slightly stronger correlations, than those who were married.

The null hypothesis was rejected which confirms that some demographic factors do significantly influence campaign success. Recommendations for future research include expanding the dataset to include variables like income and customer loyalty, adjusting sample sizes, and exploring time variables for call timing and seasonality. The insights can help tailor marketing strategies and improve the effectiveness of CD campaign outcomes.

2.0 Introduction

For my portfolio project I have chosen to dig deeper into the marketing side of the banking industry. The dataset I chose for this research comes from a bank with multiple marketing campaigns. The data included is from a Portuguese bank, which specifically focused on telemarketing calls made to customers. The variables in the dataset consists of demographics such as age, job, marital status, education, and previous campaign outcomes. These fields will be used to gain a view of how customer demographics match with specific marketing campaigns. By understanding user demographics and preferences we can predict which marketing campaign for CDs will work best. These predictive models can also identify the next product a customer might get and recommend them a campaign or offer based on that. The research from this project is geared to show how data can give insights to make data driven decisions. This would overall improve marketing strategies and increase campaign success rates.

The CD marketing campaign dataset from Kaggle has 17 total variables with 45,211 values. The data gives outcomes from the telemarketing campaigns made by the bank. The main types of variables are demographics and campaign outcome.

Focused Variables

- **Binary:**
 - Subscribed: This variable indicates whether a customer bought a certificate deposit (Yes/No).
 - Default: The variable indicates whether the customer has filed for bankruptcy and/or has credit in default (Yes/No).

- **Nominal:**

- Job: The customers role within their company. Example, like Clerk, Engineer, Services, etc.
- Marital Status: The marriage status of the customer (Single, Married, etc.).

- **Ordinal:**

- Education: The level of education the customer has (ex. primary, secondary, other).

- **Interval:**

- Age: Current age of customer.
- Duration: The telemarketing call length in seconds.

3.0 Objectives

The primary goal of this capstone project is to analyze the banking dataset to understand the impact that demographic variables, such as age, education, and marital status, have on the success rates of Certificate Deposit marketing campaigns. The specific objectives are listed below:

1. Uncover Demographic Influence: Find which demographic variables have the greatest impact of the successful outcomes for CDs.
2. Identify Key Predictors: Using descriptive statistics and correlation analysis, analyze results to find the strongest predictor to use within model.
3. Improve Predictive Models: Using the analysis from the prior objectives, apply the knowledge to the logistic regression model to improve model.
4. Provide Data-Driven Recommendations: Using the final insights, give a data driven insights to stakeholders for improving their business. Give recommendations on how to improve model and build off of it for future analysis.

4.0 Hypothesis Overview

For this project there were many business questions that could be answered using the dataset. The main concepts I focused on were demographics, key indicators, and the predictive model usage. I had to address the challenges faced by banks with improving and optimizing the success of their marketing campaigns. The below questions are the top business questions I was able to produce:

1. How can member demographics predict a successful outcome of a Certificate Deposit marketing campaign?
2. What are the key factors influencing member responses to the telemarketing campaigns?
3. Can predictive models improve the overall segmentation of marketing campaigns?

These business questions will provide the company with actionable insights for optimizing marketing strategies and improving customer acquisition efforts.

The hypothesis for this project will be related to the key indicators with the variables. There will still be a focus towards the overall success of the marketing campaigns. The research questions have helped shape the hypotheses below.

H1 (Null Hypothesis 1): Member demographics (Ex. age, education, marital status, job) do not significantly impact the success rate of CD marketing campaigns.

H2 (Alternative Hypothesis 2): Member demographics (Ex. age, education, marital status, job) significantly impact the success rate of CD marketing campaigns.

The null and alternative hypotheses will better tailor the analysis and modeling for the project. It will help determine the relationships between the variables in the dataset.

5.0 Literature Review

This article is geared towards the methods and techniques used to optimize marketing. It more supports my project because of the end goal. The analytics developed from these telemarketing campaigns will need to be optimized. The article gives a structured outline that includes customer behavior analysis with predictive models to improve targeting strategies. This research aligns closely with my project's objective of using logistic regression and decision tree models to predict telemarketing campaign success. Being able to show how predictive modeling can help identify specific variables affecting campaigns. Overall, this study helps direct the importance of using data to improve marketing on any level.

Within the customer segmentation article, I learned how similar this is to my current project. It's more tailored toward the demographics which is directly the same as my project. The article overall indicated that segmenting customers based on age, marital status, and education significantly improves campaign success rates. I'm hopeful to incorporate this study since it correlates directly with my project's hypothesis that member demographics impact the success of CD marketing campaigns. The article in general just provides me with knowledge around customer segmentation and why companies need to do it for marketing.

This article is more focused on the analysis of demographic data in telemarketing. They mainly focus on how different customer profiles respond to targeted marketing efforts. In this case study, the bank campaigns appeared to show that age and education are the two highest for influence on predicting marketing campaign responses. This is similar to my project because I am also looking into demographic data to understand which has the greatest influence. Overall,

this study does validate the use of predictive modeling and the uses for identifying the top influenced variables.

The research by Bhagyam, Reddy, and Singh (2023) explores predictive models in banking telemarketing campaigns, highlighting the effectiveness of logistic regression and decision trees in improving campaign outcomes. The authors emphasize the importance of analyzing historical campaign data to identify patterns and trends. This study supports my project's focus on using statistical models to predict customer responses and enhance segmentation strategies. By applying similar methodologies to my dataset, I aim to uncover actionable insights that can guide future marketing efforts. The findings from Bhagyam et al. reinforce the value of predictive analytics in driving successful telemarketing campaigns.

6.0 Methods

The tool used for this project will be SAS Studio for data analysis and modeling. It will be used to produce statistics and visualizations around the certificate deposit marketing dataset.

Techniques to be used:

Descriptive Statistics:

- Will create summary statistics such as mean, median, and standard deviation for all variables.
- Descriptive statistic will be used for initial view of data to understand tendencies and the overall data.

Logistic Regression:

- Will be used to predict the likelihood of customers purchasing a certificate deposit.
- Used to identify key indicators for predicting certificate deposit purchases with variables such as Age, Job, and Education.

Data Visualization:

- The visualizations will be created within SAS Studio. Using charts, graphs, and bar charts for categorical variables and histograms for numerical variables.
- These visualizations will be used to identify patterns and tendencies within the dataset.

Variable	Description	Type	Values
Age	Age of the customer	Integer	Numeric (18–95)
Job	Occupation of the customer	Nominal	Admin, Technician, Services, etc.
Marital	Marital status	Nominal	Single, Married, Divorced
Subscribed	Marketing response outcome	Binary	Yes/No
Duration	Call duration in seconds	Integer	Numeric (1–3000)
Education	Education level	Categorical	Unknown, Secondary, Primary
Default	Is user in default/bankruptcy	Binary	Yes/No
Balance	Average yearly balance	Integer	Numeric balance in eros
Housing	Housing loan	Binary	Yes/No
Loan	Personal Loan	Binary	Yes/No
Month	Last contact month of year	Categorical	Apr, May, Feb
Pdays	Number of days passed after user was last contacted	Integer	Numeric (1, 31, 54)
Poutcome	Outcome of prior marketing campaign	Categorical	Success, Failure, Other

Variable	Description	Type	Values
Contact	Communication type	Categorical	Unknown, Telephone, Cellular
Day	Last contact day of month	Integer	Numeric (1-31)

The banking industry is heavily reliant on successfully targeting and converting their members to additional products. This is a key driver for continuous revenue and membership growth. Some of the main purposes and rationale for choosing this dataset are the variety of variables, amount of data, and relevance to answering my business questions.

Relevance to business questions: This dataset directly aligns with what I envisioned for a project. The focus on the marketing side of banking and the analysis of understanding the key indicators for predicting behavior is why I choose this.

Variety of variables: With a mix of demographic, behavioral, and campaign-related variables. The dataset gives an in-depth view of potential indicators for influencing customer decisions.

Size of dataset: The dataset's large size will help with the statistical significance of the model.

7.0 Methodology

The methodology behind this project involves a variety of statistical modeling and summaries to better explore and validate the hypotheses. The main steps for methodology are data preparation, analysis, predictive modeling, and tools used. Below are the listed steps regarding methodology and tools used:

Data Preparation:

- The bank dataset has 17 variables and over 45,000 records. It will need to be cleaned in order to address missing values and outliers. The key variables I'll need to focus on is demographics like age, marital status, and education. There will also be focus on the campaign success variable since it's what I'll be using to train the model.
- There may need to be data transformations with formatting of dates and other categorical data. Some examples would be job and education. Also, I'll need to confirm fields like age are numeric.

Statistical Analysis:

- Descriptive statistics, such as mean, median, and standard deviation, will summarize the dataset, highlighting key trends and distributions.
- Chi-square tests will assess relationships between categorical variables. Some examples would be education and campaign success.

Predictive Modeling:

- Logistic regression will be the primary model to predict campaign success probabilities based on customer demographics.
- Decision tree models will provide insights into the segmentation aspect of the variables.
- Linear Regression model to understand the key indicators that correlate with the campaigns

Tools:

- SAS Studio will solely be utilized for statistical analysis, modeling, and visualization. Since it's the only resource I will be creating charts and graphs to help support or oppose the hypothesis.

The tool used for this project will be SAS Studio for data analysis and modeling. It will be used to produce statistics and visualizations around the certificate deposit marketing dataset. Descriptive statistics will create summary statistics such as mean, median, and standard deviation for all variables. Also, it will be used for initial view of data to understand tendencies and the overall data. Will be used to predict the likelihood of customers purchasing a certificate deposit. The logistic regression will be used to identify key indicators for predicting certificate deposit purchases with variables such as Age, Job, and Education. The visualizations will be created within SAS Studio. Using charts, graphs, and bar charts for categorical variables and histograms for numerical variables. These visualizations will be used to identify patterns and tendencies within the dataset.

8.0 Ethical Considerations

There are many ethical dilemmas when it comes to this capstone project. First, is data privacy since this is a bank dataset with over 45,000 customers on it. Keeping customer data safe and secure by anonymizing all personal identity information is key. To do this I have made sure the account numbers and personal information is masked. By following the regulations under the GLBA act. Lastly, transparency and documentation go hand in hand but can be an ethical consideration based on how you document and present methodologies. By documenting you will get more out of the project.

Another ethical concern is bias with the predictive models. Any demographics like, age, education, or marital status, could influence model predictions. This could then cause certain groups to get treated unequally. To reduce the risk of bias I have to regularly check the model to ensure there are no biases. If there is a bias, I need to document and see where I can tweak the model to conform to the bias. In an instance such as model bias I would encourage feedback from stakeholders as this touches on the quality and diversity of the model.

Transparency and documentation are also ethical considerations that need to be accounted for. If there is a lack of communication and documentation of the methodology or results it could cause the data to be misinterpreted. This causes confusion all around and could be resolved with a few simple steps. By documenting the steps taken to complete the model, you could understand the transformation process, model decisions, and limitations the model may have. Being transparent strengthens the model's credibility and allows individuals to audit your work.

9.0 Limitations

The dataset has 45,211 records with 17 variables, but issues such as missing or inconsistent data could impact the reliability of insights. Some categorical variables like marital status may have undefined fields labeled as unknown or null. These values would create issues within the model and would need to be cleaned in order to have a working predictive model. A cleaning process would also find any missing values or outliers within the dataset.

Using the banking data for predictive models like regression and decision trees could uncover biases in the dataset. As mentioned in the ethical dilemmas the demographic variables like age or education could cause the limitations to the predictions of campaign success. Another issue is with decision trees. They are somewhat difficult to interpret, but compared to other techniques I've used like logistic regression could cause confusion and a technical barrier between stakeholders. It would be hard for them to understand and break the barrier since they are non-technical.

This project relies only on SAS Studio for analysis and visualization. Other models I've worked on have used multiple tools so this could potentially limit the capabilities and exploration of other model techniques or visualizations. Overall, by understanding the limitations it will allow the other pieces of documentation to flow naturally.

Correlation Analysis by Marital Status:

marital=divorced

1 With Variables:	y
6 Variables:	age balance duration campaign pdays previous

Pearson Correlation Coefficients, N = 5207

	age	balance	duration	campaign	pdays	previous
y	0.12605	0.06660	0.45444	-0.06784	0.06921	0.07193
y						

marital=married

1 With Variables:	y
6 Variables:	age balance duration campaign pdays previous

Pearson Correlation Coefficients, N = 27214

	age	balance	duration	campaign	pdays	previous
y	0.10147	0.05235	0.37992	-0.06889	0.11709	0.09715
y						

marital=single

1 With Variables:	y
6 Variables:	age balance duration campaign pdays previous

Pearson Correlation Coefficients, N = 12790

	age	balance	duration	campaign	pdays	previous
y	-0.08130	0.05444	0.39793	-0.07846	0.08879	0.09348
y						

Correlation Analysis by Education Status:

education=college

1 With Variables:	y
6 Variables:	age balance duration campaign pdays previous

Pearson Correlation Coefficients, N = 13301

	age	balance	duration	campaign	pdays	previous
y	-0.00351	0.03838	0.37249	-0.08649	0.13203	0.07917
y						

education=primary

1 With Variables:	y
6 Variables:	age balance duration campaign pdays previous

Pearson Correlation Coefficients, N = 6851

	age	balance	duration	campaign	pdays	previous
y	0.15474	0.06894	0.44346	-0.05370	0.06415	0.09313
y						

education=secondary

1 With Variables:	y
6 Variables:	age balance duration campaign pdays previous

Pearson Correlation Coefficients, N = 23202

	age	balance	duration	campaign	pdays	previous
y	0.02323	0.04633	0.40585	-0.07097	0.09488	0.10675
y						

education=unknown

1 With Variables:	y
6 Variables:	age balance duration campaign pdays previous

Pearson Correlation Coefficients, N = 1857

	age	balance	duration	campaign	pdays	previous
y	0.02803	0.08868	0.33345	-0.08117	0.14718	0.13724
y						

Class Level Information		
Class	Levels	Values
education	4	college primary secondary unknown

Dimensions	
Number of Effects	2
Number of Parameters	5

Least Squares Summary				
Step	Effect Entered	Number Effects In	Number Parm's In	SBC
0	Intercept	1	1	-102623.86
1	education	2	4	-102831.26*
* Optimal Value of Criterion				

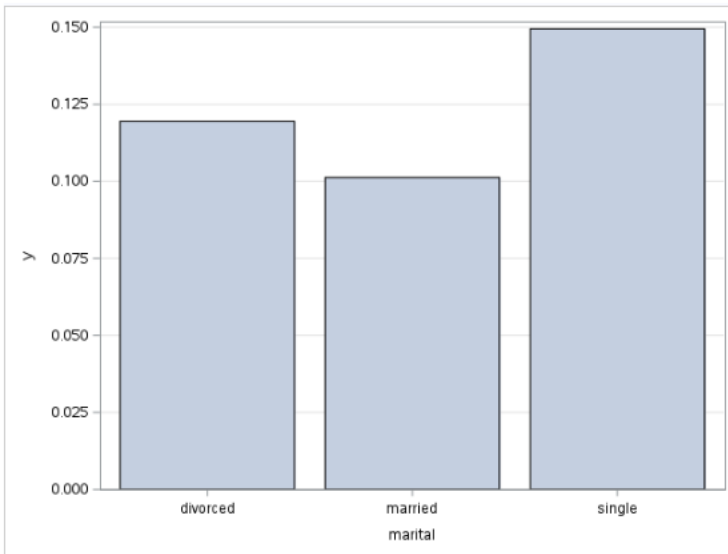
Least Squares Model (No Selection)

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	24.68065	8.22688	80.06	<.0001
Error	45207	4645.58672	0.10276		
Corrected Total	45210	4670.26737			

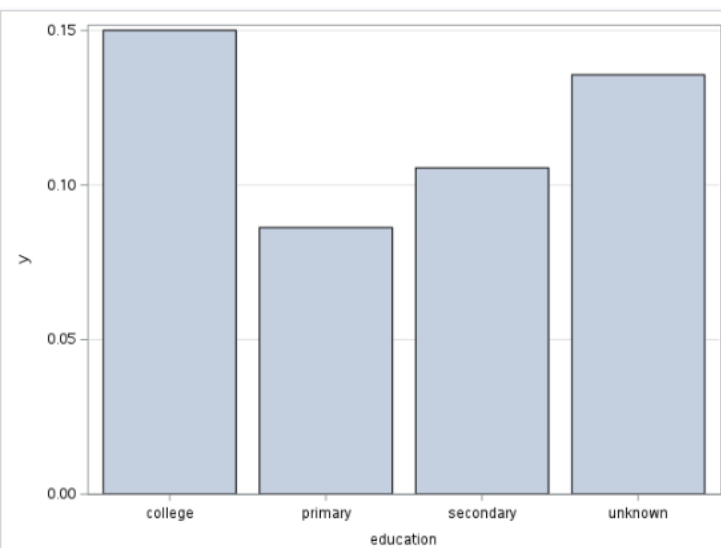
Root MSE	0.32057
Dependent Mean	0.11698
R-Square	0.0053
Adj R-Sq	0.0052
AIC	-57653
AICC	-57653
SBC	-102831

Parameter Estimates					
Parameter	DF	Estimate	Standard Error	t Value	Pr > t
Intercept	1	0.135703	0.007439	18.24	<.0001
education college	1	0.014361	0.007941	1.81	0.0705
education primary	1	-0.049438	0.008387	-5.89	<.0001
education secondary	1	-0.030108	0.007731	-3.89	<.0001
education unknown	0	0	-	-	-

Percentage of successful outcomes vs Marital Status



Percentage of successful outcomes vs Education



In my analysis, my null hypothesis states that demographic variables such as age, education, and marital status do not significantly impact the success rate of CD marketing campaigns. The goal of my analysis is to see if this null hypothesis is accepted or rejected and give the reasoning behind it. My initial analysis was of the summary statistics which gives an

overview of the variables with statistics on the mean, standard deviation, min, max, and standard error. The benefit of using this is to quickly uncover any trends or potential data discrepancies. With this dataset, the data fairly consistent and there were only a few variables that stood out. The first was Duration because of the standard deviation of 257.52 since this varies significantly as calls can last a variety of different times. The last was the Balance variable as it had negative numbers, and a significantly larger standard error compared to the other variables. This is important because it could affect the overall analysis of each variable.

The correlation analysis painted a better picture of how each variable performed relative to the overall outcome. The three highest correlation variables were Duration at .394, Balance at .052, Age at .025, and Pdays at .103. Duration was by far the highest correlation indicating it had a strong impact on the outcome of CDs which would help support the alternative hypothesis. Another thing to note was that age had a slight correlation to the outcome, but not as significant as I initially thought it would.

For the next phase of this analysis, I looked into the Marital status and Education variables to understand if there were any further correlations. The correlation results for Marital status seemed to indicate that “Divorced” individuals had a slightly stronger correlation compared to others. The slightly stronger correlation could be caused by lack of data since the sample size for divorced was around 25% compared to the other values which made up more within the dataset. For the Education correlation analysis there seemed to be a significantly higher correlation with “Primary” education compared to the other education levels. Primary education in this instance is equivalent to a high school diploma. Within Primary education the Age, Duration, and Balance had a slightly higher correlation with the outcome. The biggest

difference was with Age at .154 compared to other categories it had a much higher correlation. This finding of higher correlations with these variables further supports the alternate hypothesis and would reject the null hypothesis.

Understanding that Marital status and Education may have a correlation with the overall outcome I completed an analysis of the parameter estimates. In this analysis, there was a significant result which suggest that college education has a positive impact. While Primary and Secondary education levels could cause a negative effect on the outcome. After analyzing these results, I created a bar chart to show the significant changes in successful outcomes compared to education. The chart indicated that college was in fact the highest outcome rate which supports the other findings within the correlation analysis.

11.0 Conclusion

The analysis and findings used to evaluate the null hypothesis was rejected. This means the demographic variables do play a significant role in influencing successful outcomes. Some of the key findings came from the correlation analysis with Duration, Balance, Age, and Pdays. Duration had the strongest correlation by far at .394 which could be explained. Duration is the length of the phone call between the customer and associate. Therefore, the longer the phone call means more engagement which leads to a more positive outcome. Further analysis with Marital status and Education seemed to confirm the earlier results with the other variables. However, some of the categorical data analysis had smaller sample sizes which may have altered the results slightly. Education did emerge as the top factor, specifically with customers holding a college level education. The bar chart analysis supported these findings by illustrating a positive relationship between college education with successful outcomes.

Overall, the results suggest that demographic factors, particularly education level, significantly influence the successful outcomes for CD products. With these insights, the company could adjust and tailor its marketing strategies to its customers which would increase the likelihood of successful outcomes even more.

12.0 Recommendation

There are a few recommendations for future research that would help improve the project. This includes addressing the smaller sample sizes and looking into other variables such as Age, Occupation, etc. The foundation for this project has been built so building off it is needed to further analysis the whole dataset to overall improve the predictive model. Below are a few of the recommendations:

1. **Expand Dataset:** Include more demographic variables like income, additional products, time between purchases, and loyalty scores. This would help gain an understanding of how these factors impact the success of CDs.
2. **Balance Sample Groups:** Adjust the sample sizes accordingly, especially for marital status categories, to reduce the risk of skewed results.
3. **Seasonality Variables:** Explore periods of time, such as seasonality or time of day for calls, to understand if they influence outcomes.
4. **Targeted Marketing:** Use the insights gained from analysis to tailor marketing strategies on demographics like education to produce higher success rates.

By pursuing these recommendations, there could be even more insights gained which would positively impact this project and the predictive model used.

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